

Papers Overview

THCA (갑상선암) project 의 **평행 진행** 6 papers — claim, status, target, 핵심 figure, advisor 결정 의제, deadline 한 페이지 통합 view

6 PAPERS IN FLIGHT	I SHIP-READY (NPJ)	I HARD DEADLINE (NEURIPS 5/6)	3 DRAFTING / SUBMISSION PREP	2 STORY BRIEF 완료	~6 mo MARATHON WRITING WINDOW
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저자 · SEUNGHO COOK (1ST ACROSS ALL) 지도교수 · YU HYEONG-WON DATE · 2026-05-02
MEMORY · V17 MARATHON MODE

§ 1 · SHIP-READY / SUBMITTED

이미 제출 가능 / 제출 임박

User keyboard 만 기다리는 paper – 자료는 다 됨.

An 8-gene RAI-responsiveness biomarker outperforms BRAF V600E status in PTC

v17 KOREAN sprint K2 + Fig 8 Cox-label 정정 완료 manuscript

✓ SHIP-READY

NPJ PRECISION ONCOLOGY

2026-04-27 LAST UPDATE

핵심 CLAIM

8-gene RAI panel (SLC5A5/TPO/TG/TSHR/PAX8/NKX2-1/FOXE1/DIO1) 가 BRAF V600E status alone 대비 $\Delta AUC = +0.132$. 5-fold CV AUC 0.954.

COHORT

TCGA-THCA n=504 + MSK-IMPACT n=117 + K2 (PRJEB11591) Korean pilot n=9

SURVIVAL

Lifelines TERT⁺ Cox HR=4.33 (univariate joint 4-group), HR=0.95 multivariate stage·age·sex

MANUSCRIPT

manuscript_v6.{md,html,pdf,docx} ~12.9 K words

위치

project/submission/npj/

OUTREACH

4 outreach drafts NOT sent – user keyboard pending (memory v17_npj_ship_status)

→ [npj dashboard](#) → [START_HERE](#) → [README](#) → [Submission checklist](#) → [일반 explainer](#)

→ [20+ reviewer Q&A](#)

DIAL: A post-hoc diagnostic for subspace-aligned conditional shift under linear batch correction

v15 NeurIPS 2026 main track submission, double-blind



DEADLINE 5/4 ABSTRACT / 5/6 PAPER

NEURIPS 2026 MAIN

12 PAGES COMPILED

핵심 CLAIM

DIAL framework (Direction Identifiability After Leave-out) – 1-number AUC-based diagnostic 가 batch correction 의 class signal leak 직접 detect. THCA/ComBat 에서 $\Delta\text{DIAL} = -0.49$ leak 자체 detection (v5.1 → v5.2 self-audit).

METHOD

Synthetic 4-domain reproduction + 2 lemmas + 1 theorem (Theorem 2 status: 8/8 todos resolved → stays as theorem) + 5 experiments + 52 bibliography entries

COMPILE

pdflatex+bibtex clean, anonymisation passed, page count 12 (8 main + 4 appendix; NeurIPS main limit 10)

위치

`project/submission/v15_neurips/`

SUBMIT READY

GO verdict (submission_metadata.json), remaining: OpenReview profile + anonymous_code.zip URL replacement

FALLBACK VENUE

ICML 2027 / TMLR / AISTATS 2027

→ [FINAL PDF \(12 pages\)](#) → [Submission audit](#) → [9-day countdown](#) → [Deadline calendar](#)

→ [Live dashboard ★ NEW](#)

An 8-gene panel reveals fusion-driven, epigenetically silenced dark matter in BRAF/RAS-negative thyroid cancer

Cancer paper / Dark Matter v8 – first-pass drafts complete (00-09 manuscript files)

 DRAFTING (W1-W6)

CELL REP MED 1순위 → JCI INSIGHT + NAT COMMUN REACH → NPJ PREC ONC FALLBACK

~5,514 WORDS V1 COMPILED

핵심 CLAIM

BRAF/RAS-negative dark matter (~23%) 의 8-gene panel 기반 sub-stratification: **DM1 76.8% tyrosine-kinase-fusion+** (RET/NTRK/ALK/BRAF, OR 7.41 vs DM2) + **fusion-independent promoter hypermethylation** (TPO d=2.30, mean β 0.385 vs 0.253). **Pooled OS HR=2.53 [1.31, 4.89]** (TCGA + MSK).

COHORT

TCGA-THCA n=504 + MSK-IMPACT n=117 + Korean n=874 + cBioPortal SV n=542 + HM450 methylation n=503

5 PILLARS

I HLA forest · II GSE286332 · III Driver mRNA neutrality · IV Pan-genome robustness · V Autoimmune mechanism

MANUSCRIPT FILES

00 title · 01 abstract (153 w) · 02 outline · 03 intro (715 w) · 04 results (3250 w) · 05 figs · 06 discussion (1380 w) · 07 STAR methods · 08 cover · 09 reviewer Q&A

위치

project/manuscript_v8/ + companion references.bib 17 cites

VOICE-PROTECTED

Hook / Aim / Disc 3.1 / Limitations / Cover Para 1 / Q9 = user 키보드

→ [Paper 1 README](#) → [Abstract \(153 w\)](#) → [Outline](#) → [Reviewer Q&A pre-empt](#)

→ [Yu review dashboard \(5-pillar\)](#)

자가면역-overlap PTC 의 분자 축 (HLA-II-매개 dedifferentiation)

Phase 1 – Korean PTC+HT 의 5-Pillar evidence + § 10 mediation chain

✓ STORY BRIEF 완료 (PAPER2_BRIEF 38 PAGES)

CELL REP MED / JCI INSIGHT

5/4-6/13 WRITING MARATHON

핵심 CLAIM

한국인 PTC+HT 에서 **HLA-II Cohen d=+3.65** (saturated), 8-gene RAI d=-1.60. TCGA cross-cohort: DM2 가 Hashimoto-overlap home cluster **OR=0.20, p=6.4e-10**. Mediation: **HLA-II 140% mediated** (p=0.023), generic immune 88% NS – confounder vs mediator 정량 분리.

COHORT

TCGA n=500 + GSE286332 (Lim 2025 Dongguk n=18) + GSE213647 (Lee 2024 n=632) + K2 PRJEB11591 n=235 + Chu 2018 Han Chinese GD n=2,958

5 PILLARS

I HLA susceptibility ★ STRONG (DPB1*05:01 53.2%, pooled OR 2.16) · II in-cohort · III generalize · IV antigen-driven · V DM1 sub-B = NBNR

BRIEF DELIVERABLES

Story 38 pages + Extended 50 pages + 8 companion docs (DATA_SOURCES, COHORT_ACCESS, PROMPT_DECISION, CHANGELOG, POST_COMMIT, REFERENCES_BIB_AUDIT)

위치

project/manuscript_p2_brief/ (manuscript dir 별도 – TBD when writing 시작)

MANUSCRIPT

아직 X – 본 brief 가 advisor 미팅 자료, manuscript 본문은 6/11 W6 end 이후

→ [Story brief \(38 p\)](#) → [Extended brief \(50 p\)](#) → [Brief README](#) → [Cohort access guide](#)

→ [Data sources \(26/26 verified\)](#)

Recovered external validation and proper GSEA define DM_I/DM₂ as an actionable thyroid cancer transcriptional axis

DIAL framework methods paper – methodology + reproducibility 강조

⌚ COVER LETTER + FIGURES + TABLES + REPRODUCIBILITY 완료

BIOINFORMATICS (OXFORD OUP)

2026-04-27

핵심 CLAIM

(i) **DIAL framework** (leak-detection safeguard for batch-correction protocols) – v5.1 → v5.2 self-audit. (ii) **5-cohort validation pipeline** for DM₁/DM₂ axis recovery. (iii) Gene-ID recovery + proper preranked GSEA (100% sign-agreement with prior proxy GSEA on overlapping pathways).

PIPELINE

Deterministic seeds (random_state=42), version-pinned (sklearn 1.8.0, gseapy 1.13, scanpy 1.12.1), per-fold ComBat-seq, 8-gene RAI panel $\Delta\text{AUC}=+0.132$ vs BRAF V600E (downstream application)

SUGGESTED REVIEWERS

Aedin Culhane (U Limerick) · Lior Pachter (Caltech) · Casey Greene (UC Boulder) – all reproducibility/integration experts

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project/submission/bioinformatics/ + cover_letter.md, figures/, tables/, reproducibility/

→ Cover letter → README ★ NEW → Figures dir → Tables dir

A driver-orthogonal transcriptomic axis (DM_I/DM₂) stratifies thyroid cancer immune state and 8-gene RAI responsiveness

DM1/DM2 axis paper – mechanistic depth + cross-cohort validation 강조

COVER LETTER + FIGURES + TABLES + REPRODUCIBILITY 완료

GENOME MEDICINE

2026-04-27

핵심 CLAIM

513 TCGA-THCA primary tumor 의 unsupervised 분석으로 DM1/DM2 transcriptomic axis 정의 – **BRAF/RAS mutation 과 statistically orthogonal** (17/335 mutation-carrying tumor 가 canonical map 위반). 8-gene panel 5-fold CV AUC 0.954, Δ AUC=+0.132 vs BRAF V600E.

MULTI-OMICS

(i) bulk RNA 5 cohorts (TCGA + GSE27155 + GSE33630 + GSE29265 + GSE76039), (ii) sc-RNA 66K cells / 7 patient (GSE184362), (iii) GSEA Hallmark v2024.1.Hs (51 pathways, FDR<1e-15 inflammatory in DM1), (iv) immune-evasion (PD-L1/IDO1/HLA/CTLA-4), (v) PRISM 19Q4 4,517 compounds (MEK + HMGCR DM1-selectivity), (vi) pan-cancer transfer (LUAD/COAD/LGG/SKCM)

SUGGESTED REVIEWERS

Alex Prat (Hospital Clínic Barcelona) · Charles Perou (UNC) · Yuri Nikiforov (Pittsburgh)

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project/submission/genome_med/

→ [Cover letter](#) → [README ★ NEW](#) → [Figures dir](#) → [Tables dir](#)

§ 3 • PAPER MATRIX

6 Papers — claim × cohort × venue × deadline 통합 표

Advisor 미팅에서 한 페이지로 전체 portfolio 확인 가능.

Table 1. 6 papers × claim/cohort/venue/timeline 통합. (★ = advisor 핵심 결정 의제)

#	PAPER	핵심 CLAIM	COHORT	VENUE	STATUS	DEADLINE
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#	PAPER	핵심 CLAIM	COHORT	VENUE	STATUS	DEADLINE
0	npj 8-gene RAI	Δ AUC +0.132 vs BRAF V600E	TCGA + MSK + K2 pilot	npj Prec Onc	✓ Ship-ready	User send
N	NeurIPS DIAL	1-number leak diagnostic	Synthetic + THCA/ComBat	NeurIPS 2026 main	✓ Compiled	5/4 abstract / 5/6 paper 
1	Cancer paper v8 (Dark Matter)	DM1 76.8% fusion+ + epigenetic, HR 2.53	TCGA + MSK + Korean n=874 + meth + SV	Cell Rep Med 1순 위 → Nat Commun reach	🕒 Drafting	W1-W6 (5/4-6/13)
2	Autoimmune-overlap PTC ★	HLA-II 140% mediation	TCGA + GSE286332 + Korean	Cell Rep Med / JCI Insight	✓ Brief done · 🕒 Manuscript TBD	6/11 이후
B	Bioinformatics DIAL	Methodology + reproducibility	5-cohort thyroid	Bioinformatics OUP	🕒 Cover + figs done	Open
G	Genome Medicine DM1/DM2	Driver-orthogonal axis + drug	5-cohort + sc + PRISM + pan-cancer	Genome Medicine	🕒 Cover + figs done	Open

CROSS-PAPER BOUNDARIES (DISCUSSION Q9 의제)

Paper 0 / 1 / B / G 가 모두 8-gene panel + DM1/DM2 axis 를 share 하지만 framing 다름:

- Paper 0 (npj): Clinical biomarker (8-gene > BRAF V600E Δ AUC)
- Paper 1 (Cancer paper): Mechanism (DM1 fusion-driven epigenetic dark matter)
- Paper B (Bioinformatics): Methodology (DIAL framework + reproducibility)
- Paper G (Genome Medicine): Multi-omics integration (driver-orthogonal axis + drug + pan-cancer)
- Paper 2 (Brief): Autoimmune layer (DM2 = Hashimoto home cluster)
- Paper N (NeurIPS): ML methods (DIAL theorem)

Same reviewer pool risk: Cell Rep Med + Genome Med 같은 venue 가 두 paper 같은 reviewer 보낼 가능성.
Cross-paper separation 정당화 framing 필요 (Paper 2 advisor Q07 의제).

§ 4 • ADVISOR DECISIONS ACROSS PAPERS

통합 결정 의제 — 6 papers 전반

개별 paper Q&A 외에 cross-paper 결정 사항.

Table 2. Cross-paper decisions for advisor meeting.

#	QUESTION	관련 PAPER	본인 DEFAULT
X1	Citation correction (Chen → Chu 2018) 모든 paper 일괄 적용?	0 / 1 / 2 / B / G	일괄 grep+replace 권장 (Paper 2 Q9)
X2	Same reviewer pool risk – 어떤 venue 분산?	1 (Cell Rep Med) vs 2 (Cell Rep Med) vs G (Genome Med)	Sequential submit, Paper 0 npj 먼저 → Paper 1/G 분리
X3	4 paper 동시 submission – author bandwidth?	0/1/2/B/G/N 6 papers	Marathon 5/4-6/13 6주, NeurIPS 5/6 deadline 우선
X4	Co-corresponding 분배 – Paper 별 different policy?	0/1 advisor / 2/B/G 본인 default?	Phase 0 (Cancer) corresponding=advisor, Phase 1+ 본인 (postdoc transition) – Paper 2 Q05 의제
X5	NeurIPS 5/4 deadline urgency – 다른 paper 작업 중단?	N (deadline 9 days)	NeurIPS 우선, 다른 paper 5/6 이후 재개
X6	Bundang outreach – 어느 paper 의 Scenario A 변수?	1 + 2 + Phase 1 별도 paper	Bundang Graves' 자료 시 Paper 2 → Nat Commun reach 격상 (Paper 2 Q01)

§ 5 • TIMELINE (NEXT 6 MONTHS)

6-month writing marathon (5/4-II/4)

Marathon mode (메모리/v17_marathon_mode_post_pillar1) – 새 분석 X, manuscript writing 만.

Table 3. Marathon timeline — 6 papers parallel.

PERIOD	NEURIPS	NPJ	CANCER PAPER V8	PAPER 2	BIOINF	GENMED
5/4 (Mon)	Abstract submit AOE	—	W1 start	—	—	—
5/6 (Wed)	Full paper submit AOE	—	W1	—	—	—
5/7-5/13 (W1)	—	Outreach send	Methods / Suppl freeze	—	—	—
5/14-5/27 (W2-W3)	—	—	Results draft	분당 outreach 결과 wait	—	—

PERIOD	NEURIPS	NPJ	CANCER PAPER V8	PAPER 2	BIOINF	GENMED
5/28-6/10 (W4-W5)	—	Reviewer Q&A 대비	Discussion + Cover	Calibration 정리	Final compile	Final compile
6/11-6/13 (W6 end)	—	—	Submit Cell Rep Med	Paper 2 manuscript writing 시작	Submit OUP	Submit GenMed
7-9월	Reviewer wait	Reviewer wait	Reviewer wait	Writing M2	Reviewer wait	Reviewer wait
10-11월	Decisions / camera-ready	—	Revision	Submit	—	—

Papers Overview – THCA project master index. Generated 2026-05-02.

Brief deliverables (per paper): Paper 0 → [npj dashboard](#) · Paper N → [NeurIPS dashboard](#) · Paper 1 → [Yu review dashboard](#) + manuscript_v8/ · Paper 2 → [Story brief](#) + [Extended](#) · Paper B → [Bioinf README](#) · Paper G → [GenMed README](#)

Source memory: ~/.claude/.../memory/MEMORY.md (24 entries) · v17_marathon_mode_post_pillar1 · v17_npj_ship_status · v17_paper2_pillar1_forest_strong

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